

strip()

strip() --> remove both side space rstrip() --> remove left side space lstrip() --> remove left side space

```
In [1]: txt = " abc def ghi "  
  
txt.strip()
```

```
Out[1]: 'abc def ghi'
```

```
In [2]: txt = " abc def ghi "  
  
txt.rstrip()
```

```
Out[2]: ' abc def ghi'
```

```
In [3]: txt = " abc def ghi "  
  
txt.lstrip()
```

```
Out[3]: 'abc def ghi '
```

join list

```
In [9]: list1 = ['one', 'two', 'three', 'four']  
list2 = ['five', 'six', 'seven', 'eight']
```

```
In [10]: print(list1)  
print(list2)
```

```
['one', 'two', 'three', 'four']  
['five', 'six', 'seven', 'eight']
```

```
In [11]: list3=list1 + list2  
print(list3)
```

```
['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

append()

```
In [12]: l1=[3,5.6,'pragati',74,(6+5j),False]  
l1
```

```
Out[12]: [3, 5.6, 'pragati', 74, (6+5j), False]
```

```
In [13]: l1.append(40)  
l1
```

```
Out[13]: [3, 5.6, 'pragati', 74, (6+5j), False, 40]
```

extend()

```
In [14]: l2=[2,7,True,False,6,5.4]  
l2
```

```
Out[14]: [2, 7, True, False, 6, 5.4]
```

```
In [15]: l3=[3,2,4]  
l3
```

```
Out[15]: [3, 2, 4]
```

```
In [16]: l3.extend(l2)  
l3
```

```
Out[16]: [3, 2, 4, 2, 7, True, False, 6, 5.4]
```

```
In [17]: l3
```

```
Out[17]: [3, 2, 4, 2, 7, True, False, 6, 5.4]
```

copy()

```
In [18]: l3
```

```
Out[18]: [3, 2, 4, 2, 7, True, False, 6, 5.4]
```

```
In [19]: l2
```

```
Out[19]: [2, 7, True, False, 6, 5.4]
```

```
In [21]: l4=l3.copy()  
l4
```

```
Out[21]: [3, 2, 4, 2, 7, True, False, 6, 5.4]
```

```
In [22]: print(id(l3))      # in this, we used copy that's why id are different  
         print(id(l4))
```

```
2143775998592  
2143776190720
```

```
In [23]: l4
```

```
Out[23]: [3, 2, 4, 2, 7, True, False, 6, 5.4]
```

```
In [24]: l5=l4      # in this l4 & l5 are same ,that's why same id  
l5
```

```
Out[24]: [3, 2, 4, 2, 7, True, False, 6, 5.4]
```

```
In [25]: print(id(l4))  
         print(id(l5))
```

```
2143776190720  
2143776190720
```

```
In [26]: 14
```

```
Out[26]: [3, 2, 4, 2, 7, True, False, 6, 5.4]
```

```
In [27]: 14.clear()
```

```
In [28]: 15      # bcoz l5=l4 , in l4 we delete all the data
```

```
Out[28]: []
```

```
In [29]: 12
```

```
Out[29]: [2, 7, True, False, 6, 5.4]
```

insert(index,value)

```
In [31]: 12.insert(3,6.6)  
12
```

```
Out[31]: [2, 7, True, 6.6, False, 6, 5.4]
```

```
In [32]: 13
```

```
Out[32]: [3, 2, 4, 2, 7, True, False, 6, 5.4]
```

```
In [33]: 13.insert(2,True)  
13
```

```
Out[33]: [3, 2, True, 4, 2, 7, True, False, 6, 5.4]
```

```
In [34]: del 12
```

```
In [35]: 13
```

```
Out[35]: [3, 2, True, 4, 2, 7, True, False, 6, 5.4]
```

indexing

```
In [36]: 13[4]
```

```
Out[36]: 2
```

```
In [37]: 13[-5]
```

```
Out[37]: 7
```

```
In [134... t4=[3,4,6,3.4,'tejal','pragati']  
t4
```

```
Out[134... [3, 4, 6, 3.4, 'tejal', 'pragati']
```

```
In [138... t4[4][3]
```

```
Out[138... 'a'
```

```
In [140... t4[5][0]
```

```
Out[140... 'p'
```

```
In [38]: 13
```

```
Out[38]: [3, 2, True, 4, 2, 7, True, False, 6, 5.4]
```

slicing

```
In [41]: 13[0:3]
```

```
Out[41]: [3, 2, True]
```

```
In [42]: 13[-5:-2]
```

```
Out[42]: [7, True, False]
```

```
In [43]: 13
```

```
Out[43]: [3, 2, True, 4, 2, 7, True, False, 6, 5.4]
```

```
In [44]: 13[3:-1]
```

```
Out[44]: [4, 2, 7, True, False, 6]
```

```
In [45]: 13[:]
```

```
Out[45]: [3, 2, True, 4, 2, 7, True, False, 6, 5.4]
```

```
In [46]: 13
```

```
Out[46]: [3, 2, True, 4, 2, 7, True, False, 6, 5.4]
```

```
In [47]: 13[: : 2]
```

```
Out[47]: [3, True, 2, True, 6]
```

```
In [53]: 13[6:-2:1]
```

```
Out[53]: [True, False]
```

```
In [54]: 13[: :-1]
```

```
Out[54]: [5.4, 6, False, True, 7, 2, 4, True, 2, 3]
```

nested list -list inside list

```
In [57]: l1=[2,4,6.6,(6+7j),True,False,3,[4,'tejal']]  
l1
```

```
Out[57]: [2, 4, 6.6, (6+7j), True, False, 3, [4, 'tejal']]
```

```
In [58]: len(l1)
```

```
Out[58]: 8
```

```
In [59]: l1
```

```
Out[59]: [2, 4, 6.6, (6+7j), True, False, 3, [4, 'tejal']]
```

```
In [ ]: # remove 3 kel hot
```

```
In [87]: l4=[2, 4, 6.6, (6+7j), True, False, [4, 'tejal']]  
l4
```

```
Out[87]: [2, 4, 6.6, (6+7j), True, False, [4, 'tejal']]
```

remove(value)

```
In [88]: l4.remove(6.6)  
l4
```

```
Out[88]: [2, 4, (6+7j), True, False, [4, 'tejal']]
```

```
In [68]: l1
```

```
Out[68]: [2, 4, (6+7j), True, False, [4, 'tejal']]
```

```
In [81]: l2=[3,3,2,5,3,5,6,7,7,8,8]  
l2
```

```
Out[81]: [3, 3, 2, 5, 3, 5, 6, 7, 7, 8, 8]
```

count(value)

```
In [82]: l2.count(3)
```

```
Out[82]: 3
```

```
In [89]: l2.count(7)
```

```
Out[89]: 2
```

index(value)

```
In [90]: l2
```

```
Out[90]: [3, 3, 2, 5, 3, 5, 6, 7, 7, 8, 8]
```

```
In [93]: l2.index(5)
```

```
Out[93]: 3
```

pop()

```
In [94]: l1
```

```
Out[94]: [2, 4, (6+7j), True, False, [4, 'tejal']]
```

```
In [95]: l1.pop()
```

```
Out[95]: [4, 'tejal']
```

```
In [97]: l1
```

```
Out[97]: [2, 4, (6+7j), True, False]
```

```
In [98]: l2
```

```
Out[98]: [3, 3, 2, 5, 3, 5, 6, 7, 7, 8, 8]
```

```
In [99]: l3=[4,8.4,(3+6j),True,False,9,'Tejal']  
l3
```

```
Out[99]: [4, 8.4, (3+6j), True, False, 9, 'Tejal']
```

pop(index)

```
In [100... l3.pop(3)
```

```
Out[100... True
```

reverse()

```
In [101... l3
```

```
Out[101... [4, 8.4, (3+6j), False, 9, 'Tejal']
```

```
In [102... l3.reverse()  
l3
```

```
Out[102... ['Tejal', 9, False, (3+6j), 8.4, 4]
```

sort()

in sort same datatype or int,float,bool can consider

```
In [103... 13

Out[103... ['Tejal', 9, False, (3+6j), 8.4, 4]
```

```
In [110... l2=[1,1,3.4,6,8,True,False]
l2

Out[110... [1, 1, 3.4, 6, 8, True, False]
```

```
In [112... l2.sort()
l2

Out[112... [False, 1, 1, True, 3.4, 6, 8]
```

```
In [113... l2

Out[113... [False, 1, 1, True, 3.4, 6, 8]
```

```
In [114... l2.sort(reverse=True)
l2

Out[114... [8, 6, 3.4, 1, 1, True, False]
```

tunning

```
In [115... l=[1,3,5,6.6]
l
```

```
Out[115... [1, 3, 5, 6.6]
```

```
In [119... l.sort()
l          # parameter tunning
```

```
Out[119... [1, 3, 5, 6.6]
```

```
In [120... l.sort(reverse=True)
l          # hyperparameter tunning
```

```
Out[120... [6.6, 5, 3, 1]
```

list membership

in not in

```
In [142... l2
```

```
Out[142... [8, 6, 3.4, 1, 1, True, False]
```

```
In [143... 3.4 in l2
```

```
Out[143... True
```

```
In [144... 10 in l2
```

Out[144...] False

In [145...] `10 not in 12`

Out[145...] True

In [146...] `5 not in 12`

Out[146...] True

all - all value must be true , 0=False

any - any 1 value true

In [147...] `12`

Out[147...] `[8, 6, 3.4, 1, 1, True, False]`

In [148...] `any(12)`

Out[148...] True

In [150...] `all(12)` # 3.4

Out[150...] False

In [152...] `l=[1,7,5,(4+5j),'tejal']`
`1`

Out[152...] `[1, 7, 5, (4+5j), 'tejal']`

In [153...] `any(1)`

Out[153...] True

In [155...] `l=[4,8,(4+6j),True]`
`1`

Out[155...] `[4, 8, (4+6j), True]`

In [156...] `any(1)`

Out[156...] True

In [157...] `all(1)`

Out[157...] True

In [158...] `l=[5,8,0,5,(5+8j)]`
`1`

Out[158...] `[5, 8, 0, 5, (5+8j)]`

In [159... `any(1)`

Out[159... `True`

In [161... `all(1)` *#bcoz of 0*

Out[161... `False`

In [162... `t=[5,False,1.5,(6+7j),4,'teajal']`
`t`

Out[162... `[5, False, 1.5, (6+7j), 4, 'teajal']`

In [163... `any(t)`

Out[163... `True`

In [164... `all(t)` *# bcoz of False*

Out[164... `False`

In []: